

## Project Design Phase-II

### Solution Requirements (Functional & Non-functional)

|               |                                                    |
|---------------|----------------------------------------------------|
| Date          | 13 July 2026                                       |
| Team ID       | [To be filled by student/team]                     |
| Project Name  | Voice-Based Concept Understanding Analyzer (VBCUA) |
| Maximum Marks | 4 Marks                                            |

#### Functional Requirements:

Following are the functional requirements of the proposed solution.

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task)                                                                                                                                                                                                                                                                                                |
|--------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1   | Audio Upload                  | <ul style="list-style-type: none"><li>Upload a student's spoken response as an audio file (WAV, MP3, or M4A).</li><li>Validate the uploaded file type against supported formats.</li><li>Display the uploaded file name and confirmation to the user.</li></ul>                                                                   |
| FR-2   | Speech-to-Text Transcription  | <ul style="list-style-type: none"><li>Transcribe the uploaded audio file to text using the OpenAI Whisper "base" model.</li><li>Cache the loaded Whisper model in memory to avoid reloading on repeated use.</li><li>Handle transcription failures (missing file, model load failure) with a descriptive error message.</li></ul> |
| FR-3   | Reference Answer Management   | <ul style="list-style-type: none"><li>Load the ground-truth reference answer from a stored text file.</li><li>Validate that the reference answer file exists and is not empty.</li><li>Report a warning if the reference answer cannot be loaded.</li></ul>                                                                       |
| FR-4   | Semantic Similarity Analysis  | <ul style="list-style-type: none"><li>Generate sentence embeddings for the transcript and the reference answer using Sentence Transformers (all-MiniLM-L6-v2).</li><li>Compute the cosine similarity between the two embeddings.</li><li>Clamp and return a normalised similarity score between 0.0 and 1.0.</li></ul>            |
| FR-5   | Concept Understanding Scoring | <ul style="list-style-type: none"><li>Convert the similarity score into a percentage-based concept understanding score.</li><li>Assign a qualitative grade tier (Excellent / Good / Fair / Needs Improvement) based on the score.</li><li>Generate descriptive, grade-specific feedback text.</li></ul>                           |
| FR-6   | Results Display               | <ul style="list-style-type: none"><li>Display the transcribed text in a read-only panel.</li><li>Display the similarity score as a percentage metric.</li><li>Display a colour-coded grade badge corresponding to the assessment tier.</li><li>Display the generated feedback message.</li></ul>                                  |
| FR-7   | PDF Report Generation         | <ul style="list-style-type: none"><li>Generate a PDF assessment report containing the transcript, similarity score, grade, feedback, and generation timestamp.</li><li>Validate all inputs before report generation and raise descriptive errors on failure.</li></ul>                                                            |

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task)                                                                                                                                                                                                                   |
|--------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                               | <ul style="list-style-type: none"> <li>Save the generated report to the reports directory.</li> </ul>                                                                                                                                                |
| FR-8   | <b>Report Download</b>        | <ul style="list-style-type: none"> <li>Provide a download button for the generated PDF report.</li> <li>Disable the download action when no report is available.</li> <li>Display a warning if the report file cannot be located on disk.</li> </ul> |

### Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

| NFR No. | Non-Functional Requirement | Description                                                                                                                                                                                                                                                                                                             |
|---------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-1   | <b>Usability</b>           | The application provides a single-page Streamlit interface with a clearly labelled upload area, format guidance, and immediate visual feedback (success, warning, and error messages), allowing users with minimal technical expertise to operate it without training.                                                  |
| NFR-2   | <b>Security</b>            | The application restricts uploads to a defined set of audio file types and validates all inputs before processing. Since the current version is intended for local or classroom-scale educational use, it does not implement user authentication or file encryption; access control is limited to the host environment. |
| NFR-3   | <b>Reliability</b>         | The system anticipates and gracefully handles predictable failure conditions, including missing audio files, empty transcripts, missing reference answers, and failed PDF generation, by catching exceptions and presenting descriptive warnings instead of crashing.                                                   |
| NFR-4   | <b>Performance</b>         | The Whisper and Sentence Transformer models are loaded once and cached in memory for the lifetime of the application process, avoiding repeated model-loading overhead across analyses within the same session.                                                                                                         |
| NFR-5   | <b>Availability</b>        | The application runs as a single Streamlit process on the host machine. Availability is tied to that host remaining online; no redundant or cloud-hosted deployment is currently implemented.                                                                                                                           |
| NFR-6   | <b>Scalability</b>         | The current architecture relies on Streamlit session state and is designed for single-user, single-session usage. Supporting concurrent multi-user or classroom-wide usage would require additional infrastructure (e.g., cloud deployment, persistent database-backed sessions), which is identified as future scope.  |